Received: 23 August 2023 | Accepted: 30 April 2024

DOI: 10.1002/mar.22025

RESEARCH ARTICLE

Psychology & Marketing WILEY

From ugly to attractive: Leveraging anthropomorphism to increase demand for irregular-appearing produce

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Abstract

Recognizing that the waste of imperfect produce contributes to the global environmental crisis, the authors conduct three studies examining whether retailers can use anthropomorphizing marketing techniques to make irregular-appearing produce more attractive and increase purchase intentions. Based on the exemplar model theory, Study 1 shows that when retailers place googly eyes on pictures of irregular-appearing produce, consumers judge the product according to multiple exemplars as they do when they evaluate humans. The multiple esthetic cues cause them to perceive irregular produce as more attractive. Study 2 replicates Study 1 by using human names as the anthropomorphic cues, demonstrating that anthropomorphism can increase purchase intentions toward irregular-appearing produce. Study 3 further shows that the anthropomorphism effects hold for irregular-appearing produce from corporate farms and not from local farms. The differences occur because consumers expect corporate farms to conform to standardized esthetic norms but expect local farms to market irregular-appearing produce. When expectations are aligned, anthropomorphism is more likely to cause consumers to diversify esthetic cues, but not when expectations are disassociated. The article concludes with suggestions for promotion strategies that use anthropomorphizing to change attitudes and increase purchase intentions toward irregular-appearing produce.

KEYWORDS
anthropomorphism, consumer choice, esthetic appeal, exemplar model, food waste

1 | INTRODUCTION

Each year, 72 billion pounds of fruits and vegetables produced in the United States go straight from the farm to incinerators or landfills because the produce is too small, asymmetrical, or scarred to meet cosmetic standards (FAO, IFAD, UNICEF, WFP & WHO, 2022; Feeding America, 2017; Gunders & Bloom, 2017). The economic and environmental impacts are significant. The water, energy, and land resources used in producing produce are further wasted, contributing to deforestation and water scarcity (Gunders & Bloom, 2017; Kibler et al., 2018; Perrone & Hornberger, 2014). As produce decomposes, landfills emit methane, a potent greenhouse gas that contributes to climate change (Nordin et al., 2020).

In 2015, the United Nations established sustainable development goals to "halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including postharvest losses" by 2030 (United Nations, 2015). To persuade consumers to purchase imperfect produce, marketers have implemented various strategies such as offering discounts or emphasizing environmental benefits (Grewal et al., 2019; Qi et al., 2022). For instance, campaigns such as "Ugly Fruit & Veg" or "Imperfect Produce" were instituted to increase awareness that irregular produce is equally nutritious. Nevertheless, we lack research showing whether such efforts prevent consumers from adhering to their preferences for stereotypical, standardized produce.

This paper introduces a novel approach for altering consumer perceptions. We suggest that retailers can counteract negative evaluations, increase attraction, and enhance purchase intentions if they anthropomorphize irregular-appearing produce. Kahneman and Miller's (1986) exemplar theory explains that people tend to hold to simple, objective, singular, stereotypical exemplars and are repulsed by abnormalities. For example, shoppers are likely to reject a three-headed strawberry because it violates the single-headed basic standard. However, the exemplar theory also suggests that people often rely on multiple exemplars to develop a more comprehensive understanding of an object or event, and the choice of specific exemplars can be context-dependent (Hilton & Von Hippel, 1996; Smith & Zárate, 1992). Based on those understandings, we contend that when retailers give imperfect produce human features, consumers will use complex and subjective criteria based on diverse and varying exemplars, analogous to the approach they use to evaluate strangers. This suggests that anthropomorphism permits irregular-appearing produce to be evaluated based on various esthetic cues (Fournier & Alvarez, 2012), rather than solely on its non-conforming appearance. For example, the three-headed strawberry with human names might seem attractive and possessing positive characteristics. As a result, when an irregular-appearing product is anthropomorphized, consumers will increase their purchase intentions.

Furthermore, we propose that the effect depends on the type of farm where the produce is grown. Specifically, anthropomorphizing may be more effective when the irregular-appearing produce comes from large corporate farms because consumers expect corporate farms to produce esthetically pleasing standardized produce. The large corporate farm thus reinforces consumers' salient abnormal perceptions of an irregular-appearing product, and anthropomorphism may retain the intended effect of diversifying its esthetic cues and reducing negative consumer outcomes. Conversely, if the irregular-appearing produce is grown in small local farms, anthropomorphizing may fail because consumers are more likely to expect substandard esthetic norms.

In this article, we report the results of two studies in which we examined the effects of assigning human names to irregular produce to increase purchase intentions.

2 | THEORETICAL BACKGROUND

2.1 | Esthetic biases, ugly produce, and food waste

"Beauty-is-good" biases are strongly responsible for the prevalence of food waste (Castagna et al., 2021). That is, consumers tend to reject so-called "ugly foods" that violate standard, stereotypical views about ideal product appearance (Hartmann et al., 2021), resulting in constant waste of vast amounts of perfectly edible and nutritious food, exacerbating food insecurity and damaging the environment (FAO, IFAD, UNICEF, WFP & WHO, 2022). The issue must be addressed considering the crucial need for environmental sustainability and social responsibility (Diwanji et al., 2024; Ko & Phua, 2024).

Why do consumers adhere so tightly to esthetics when judging produce? When consumers evaluate most products or objects, they consider functionalities, quality, value, and even neural responses (Bloch, 1995; Creusen et al., 2010; Patrick & Hagtvedt, 2011; Reimann et al., 2010) based on various esthetic visual principles such as size (Folkes & Matta, 2004), balance (Orth & Malkewitz, 2008), proportion (Raghubir & Greenleaf, 2006), color (Garber et al., 2000), complexity (Schoormans & Robben, 1997), and unity (Veryzer & Hutchinson, 1998). Despite some exceptions (Wu et al., 2017), studies have shown that consumers expect products that have esthetically pleasing appearance to be tastier (Schnurr, 2019) and more pleasurable to purchase (Townsend & Sood, 2012). Obviously, marketers know that product design is critical for driving sales (Bloch, 1995; Khan et al., 2024).

Visual esthetic perfection also critically impacts consumer judgments and choices regarding food products (Griffin & Langlois, 2006; Koo et al., 2019). Blemish-free produce that perfectly matches standard appearance earns a "beauty premium" (Bunn et al., 1990), even though imperfect foods have the same quality, expiration dates, packaging, and safety assurances (Castagna et al., 2021; Griffin & Langlois, 2006). As such, appearance strongly affects quality assessment, both subconsciously and consciously (Qi et al., 2022). That is, food products are wasted because of "esthetic bias" driving consumers to perceive irregular-appearing produce as risky (Castagna et al., 2021). Indeed, 27 of 53 studies reviewed in the last 15 years found that abnormal appearance is the strongest reason for produce rejection (Hartmann et al., 2021).

Obviously, esthetic bias and the beauty premium contribute to retail and household food waste, ultimately leading to environmental degradation. Therefore, the world imperatively needs rigorous research that develops and tests strategies for changing consumer perceptions and motivating them to purchase and consume produce perceived as "ugly" because it deviates from stereotypical esthetic visual norms by being irregular, imperfect, misshapen, blemished, and abnormal (Aschemann-Witzel et al., 2018; Jaeger et al., 2016; McCarthy & Liu, 2017).

For our analysis here, we define ugly or abnormal produce as produce that differs significantly from typical categorical expectations regarding shape and appearance. For instance, shoppers may reject a fresh, vividly red, three-headed strawberry simply because it deviates from standard expectations that strawberries have single heads. Produce may be rejected because it appears to deviate from numerous expectations, but rather than focus on a continuous spectrum, we primarily focus on a binary classification in which produce is labeled as either irregular or regular in appearance.

2.2 | An exemplar perspective

The exemplar theory (Kahneman & Miller, 1986) is a useful perspective for understanding why consumers reject irregular produce and why the beauty premium overrides rational and normative decision-making. As a psychological framework, the theory explains that individuals cognitively categorize objects, concepts, and experiences based on how well the entities resemble and compare to examples, prototypes, or previous examples within the category rather than depending on a fixed set of regulations or a single prototype (Kahneman & Miller, 1986; Mao & Krishnan, 2006). Individuals react more positively when objects, concepts, and experiences align with and can be categorized as belonging to specific exemplars. Conversely, they have disproportionally negative reactions to objects that deviate from the standardized esthetic norms.

In our context, visual esthetics pertain to consumer perceptions regarding whether produce adheres to normal/regular shapes or perfect standards based on similar attributes. For example, consumers who are purchasing apples will reject asymmetrical samples that deviate from the exemplar in which perfect apples are symmetrically round. Similarly, consumers purchasing strawberries are likely to consider single-headed strawberries to be the exemplar for the regular strawberry category and are likely to reject three-headed strawberries that deviate from the established standard. Basically, visual esthetics signal quality and safety, while visual deviations signal abnormality, even when consumers have access to objective information assuring them of product quality (Crolic et al., 2019). That is, even when consumers know that irregularly shaped produce is safe, their low "deformation thresholds" (Di Muro et al., 2016) drive them to continue abhorring even moderately asymmetrical and slightly blemished produce (de Hooge et al., 2017; Loebnitz & Grunert, 2015; Lund et al., 2006; Symmank et al., 2018).

Researchers have taken two broad approaches to examine consumer resistance to irregular produce. The first approach is to identify barriers associated with socio-demographics, attitudes, context, and knowledge and information seeking. Demographically, most resistance comes from consumers who are female (Aschemann-Witzel et al., 2018), seniors (Cicatiello et al., 2019; de Hooge et al., 2018; Van Giesen & de Hooge, 2019), educated at low levels, and members of small households (Aschemann-Witzel et al., 2017). Knowledge and information seeking barriers are found among consumers who know little about food production and are unlikely to try to learn about irregular produce (Di Muro et al., 2016). Attitude barriers include preconceived negativity, low environmental awareness (Van Giesen & de Hooge, 2019; Wong et al., 2018), and doubt about the quality, naturalness, taste, and nutrition associated with irregular produce (Aschemann-Witzel et al., 2017). Context-related factors include discounted prices (Aschemann-Witzel et al., 2017; Grewal et al., 2019), availability (Di Muro et al., 2016), and cognitive demands (White et al., 2016).

A second approach examines intervention strategies for making consumers willing to purchase ugly foods. Suggestions include promoting irregular-appearing produce as convenience food (Aschemann-Witzel et al., 2018), using sustainable packaging (Di Muro et al., 2016; Tu et al., 2018), offering price discounts (Wang et al., 2016), varying price discount levels according to specific flaws (de Hooge et al., 2017), educating the public through communications (Van Giesen & de Hooge, 2019), and changing production processes (Yue et al., 2009). Particularly emphasized are messages that help consumers form more positive self-perceptions (Grewal et al., 2019), highlight the naturalness and quality of irregular-appearing produce (Van Giesen & de Hooge, 2019), promote sustainability, and advertise corporate social responsibility (White et al., 2016). In this article, we propose anthropomorphism as another intervention strategy.

2.3 | Anthropomorphism as an intervention to promote ugly produce consumption

Marketers commonly use anthropomorphism strategies in which they attribute human characteristics to nonhuman entities (Aggarwal and McGill 2012; Epley et al., 2007). The technique accesses the human schema by causing humans to perceive nonhuman objects as similar to themselves (Kim & McGill, 2011). Consumers tend to perceive products as endearing and likable when marketers depict them as having human-like features such as faces, hands, and names (Aggarwal and McGill 2012). For example, Volkswagen New Beetle cars are designed with round headlights and taillights that look like cheerful smiling faces. Products packaged to resemble human bodies tend to activate personal connections (De Bondt et al., 2018). Furthermore, anthropomorphic cues cause consumers to make warm (Zhou et al., 2019), compassionate (Koo et al., 2019), and tender (Wang et al., 2017) product evaluations.

As such, anthropomorphism is widely accepted as positively impacting consumer perceptions based on the halo effect in which consumers transmit humanness evaluations to nonhuman entities. However, we have reasons for rejecting oversimplified assumptions that adding anthropomorphized cues to irregular-appearing produce can automatically increase appreciation and dilute negative impressions simply. One reason for caution is that studies on the "uncanny valley effect" report that anthropomorphized objects can be repellent (Seyama & Nagayama, 2007; Strait et al., 2015), because the inconsistency with reality evokes cold, eerie, unfamiliar feelings (Chattopadhyay & MacDorman, 2016). A second concern is that anthropomorphizing was shown to increase perceptions of warmth but to decrease liking (Kim et al., 2019). Thus, mixed findings suggest that consumer responses to anthropomorphized products involve more complicated processes.

We contend that anthropomorphic cues do more than evoke positive associations. Rather, they can broaden the range of esthetic criteria used to form impressions about irregular produce. The exemplar theory suggests that people rely on multiple exemplars to comprehensively understand categories, depending on context (Hilton & Von Hippel, 1996; Smith & Zárate, 1992). Human beauty is a multidimensional construct that defies conceptualizations of attractiveness (Langlois & Roggman, 1990; Rhodes, 2006). For example, female actors such as Margot Robbie, Scarlett Johansson, and Emma Stone are considered to be at the higher end of the attractiveness spectrum although they have dissimilar physical features and styles of attractiveness, including elegance, exoticism, sexiness, and nobility. Consequently, anthropomorphic cues may create complex and nuanced perceptions. In our research context, this implies that as humans do when they evaluate others, anthropomorphism may enable consumers to form judgments about irregular produce according to multidimensional standards, which reconstructs their views of stereotypical shapes.

Drawing from the exemplar theory, we thus argue that consumer disinclination to misshapen produce stems from their deviation from the prototypical shape learned through prior experiences. However, the introduction of anthropomorphism enables misshapen produce to be assessed based on a broader range of esthetic cues, transcending mere prototypicality. Specifically, consumers have instantly favorable judgments about regular produce because it aligns with the primary exemplar and needs no anthropomorphizing. In contrast, irregular-appearing produce, which deviates from the primary exemplar, can become more appealing when anthropomorphized. This is because consumers less rely on its appearance and consider other esthetic dimensions in forming judgment. Considering that "beauty is in the eye of the beholder," they may be even able to perceive anthropomorphized irregular produce as possessing uniquely positive characteristics and being attractive. This shift in perception reduces the negative attributes typically associated with its abnormal appearance, making consumers more open to trying new and unconventional choices. Our theorization is grounded on the notion that exemplars are readily accessible mental presentations, and individuals can swiftly change judgments when their exemplars change through anthropomorphism (Kahneman & Miller, 1986).

In this regard, we anticipate that increased attractiveness will explain the positive impact of anthropomorphism on consumer purchases of irregular-appearing produce. Existing literature suggests that individuals perceive objects as more attractive when they can identify and engage with multiple esthetic cues (Hoegg & Alba, 2018; Kallio, 2003). Attractiveness is a subjective, cognitive, and emotional construct associated with favorable responses (Eagly et al., 1991). Attractiveness perceptions rise when objects are esthetically pleasing and interest arousing (Mehrabian & Blum, 2018). When viewers encounter atypical produce, they typically form ugliness perceptions due to lower attractiveness. In this case, anthropomorphism should diversify their exemplars, enhance attractiveness, and increase purchase intent. Therefore, we propose that attractiveness acts as a mediator when consumers decide whether to purchase anthropomorphized irregular-appearing produce. Formally, we hypothesize:

H1: Anthropomorphic cues will (will not) increase attractiveness perceptions toward irregular-appearing (regular) produce.

H2: Anthropomorphic cues will (will not) increase purchase intentions toward irregular-appearing (regular) produce.

2.4 | Farm type as moderator

The exemplar model explains that dominant exemplars influence perceptions of irregular produce. To fully understand how marketers can use anthropomorphic cues to enhance purchase intentions and ultimately reduce food waste, we suggest that *farm type* is a specific contextual factor that may act as a moderator determining whether anthropomorphizing through naming has positive or negative effects. That is, we assume that farm type may shape consumers' perceptions of produce appearance as a dominant exemplar, thereby potentially impacting the effectiveness of anthropomorphism. Following a moderation process approach (Bullock et al., 2010), we provide additional evidence supporting our exemplar perspective in elucidating consumers' reluctance to purchasing irregular-appearing produce. This approach transcends the limitations of bias-prone measurements commonly used in previous studies.

Farms are generally categorized as either large corporate farms or small local farms. They have different production practices, distribution and supply chains, and branding and advertising strategies. To clarify, we define corporate farms as more industrial, characterized by larger acres, greater produce per acre, a higher number of employees, and higher annual revenue when compared to local farms. Based on the lay theory that corporate farms prioritize uniform and esthetically appealing produce, while local farms prioritize taste and freshness (Hinrichs, 2003), we posit that anthropomorphizing through naming will be effective in promoting the consumption of irregular produce from corporate farms but not from local farms. Corporate farms often use mechanized systems to regulate product growth, resulting in more uniform and esthetically pleasing produce. Their supplies must be uniform because they generally go to large retailers and supermarkets that often cull or discard produce that fails to meet their strict standards about appearance. Furthermore, corporate farms have robust branding and advertising campaigns that emphasize quality and uniformity. In contrast, local farms lack the technological resources to regulate growth. Instead, they tend to produce irregular and often irregular-appearing products sold directly to consumers or smaller markets that operate under looser standards. Consequently, consumers tend to expect corporate farms to provide more esthetically normative produce that has regular appearance. Although consumers may find abnormal produce to be highly noticeable, they may view it as conflicting with their high expectations only when it comes from corporate farms.

Recall that anthropomorphic naming draws consumer attention to esthetic cues beyond the appearance of irregular produce, but consumers must first perceive that irregularity violates the exemplar rule (Dasgupta & Greenwald, 2001; Wittenbrink et al., 2001). We suggest that only irregular-appearing produce from corporate farms will appear to violate the exemplar rule, so that naming is essential to mitigate negative evaluations. In contrast, consumers may have more lenient esthetic norms for produce from local farms and can more easily overlook abnormality. Thus anthropomorphic naming may have limited effectiveness for produce from small local farms. Taken together, we hypothesized that:

H3: Anthropomorphizing (vs. not anthropomorphizing) irregular-appearing produce will increase willingness to purchase the produce through higher attractiveness perceptions, but this effect is conditional on the farm type. The positive impact of anthropomorphism for irregular-appearing produce through increased attractiveness perceptions will be significant only when the produce is sourced from corporate farms, but it will not have the same effect for produce from local farms.

3 | PRETEST

To test our hypotheses, we selected pictures showing six pairs of regular-appearing and irregular-appearing lemons, strawberries, peaches, eggplants, tomatoes, and carrots (see Appendix). We conducted a separate pretest to confirm that participants would perceive the product appearance as intended. Sixty-six undergraduate students from a Northeastern U.S. Uuniversity evaluated the produce pairs according to three items on seven-point bipolar scales: (1) irregular-appearing/regular-appearing, (2) ugly/pretty, and (3) misshapen/well-formed. Their responses were averaged to create a composite measure.

As expected, a series of paired-sampled t-tests showed that participants perceived the regular-appearing produce as more regular-appearing (vs. irregular-appearing), pretty (vs. ugly), and well-formed (vs. misshapen), compared to the irregular-appearing produce, across all listed fruits and vegetables: lemon ($M_{regular} = 6.73$ vs. $M_{irregular} = 1.71$; $t\ (65) = 30.48$, $p < 0.001$), peach ($M_{regular} = 6.62$ vs. $M_{irregular} = 2.23$; $t\ (65) = 21.12$, $p < 0.001$), strawberry ($M_{regular} = 6.78$ vs. $M_{irregular} = 1.98$; $t\ (65) = 24.71$, $p < 0.001$), eggplants ($M_{regular} = 6.67$ vs. $M_{irregular} = 1.77$; $t\ (65) = 26.44$, $p < 0.001$), tomatoes ($M_{regular} = 6.73$ vs. $M_{irregular} = 2.77$; $t\ (65) = 15.54$, $p < 0.001$), and carrot ($M_{regular} = 6.70$ vs. $M_{irregular} = 1.72$; $t\ (65) = 26.17$, $p < 0.001$).

3.1 | STUDY 1

We conducted Study 1 to test H1 predicting that anthropomorphizing misshapen, irregular-appearing produce will make study participants perceive it as more attractive, but that anthropomorphizing will fail to enhance attractiveness perceptions for regular-appearing produce. We used eggplants as the target stimulus and manipulated anthropomorphism by using googly eyes as a visual cue, which is one of the most common cues that scholars and marketers have used to humanize objects (Kim & McGill, 2011; Kim et al., 2016).

3.2 | Method

Study 1 utilized a 2 (produce shape: regular-appearing vs. irregular-appearing) × 2 (anthropomorphism: googly eyes vs. no-eyes) between-subjects design. We recruited 167 U.S. participants from Prolific, an online platform that researchers use to gain access to broad populations from diverse backgrounds and demographics (Palan & Schitter, 2018). Participants were paid $1.58 for viewing advertisements and answering questions about their responses.

We created four advertisements for *Del Campo*, a fictitious farm brand supplying fresh vegetables to grocery stores. In a separate pretest, we asked 96 undergraduate students from a northeastern U.S. university whether they were familiar with a farm named *Del Campo*. The results indicated limited brand recognition for the name *Del Campo*: eighty-one participants (84%) were unaware of the brand; eleven (11%) were unsure, three (3%) thought they had heard of it.

Participants in the regular-appearing produce condition viewed ads depicting symmetrical, balanced, regular-appearing eggplants, while those in the irregular-appearing produce condition saw ads featuring abnormal, ugly, and misshapen produce. Consumers often perceive irregular-appearing produce as rotten, spoiled, moldy, or harmful, leading to disgust, aversion, and health concerns (Rozin et al., 1986). To ensure that participants based their judgments solely on the abnormality of produce appearance, the ad content uniformly emphasized nutritional benefits. For example, the ad message read, "Eggplant offers a host of health benefits. It contains antioxidants, is fat-free, saturated fat-free, sodium-free, cholesterol-free, and low in calories." We also portrayed *Del Campo* as a farm that provides special care for its vegetables and provides nutritional information for its customers.

We manipulated anthropomorphism by adding googly eyes to pictures of eggplants. Eyes, often called "windows to the soul," hold profound spiritual symbolism. Social signal theory explains that gaze behavior is a social signal (Minato et al., 2004). As such, when humans interact, they typically focus mostly on one another's eyes (Bindemann et al., 2009). Empirical studies in the prosocial domain show that both actual eye images and eye-like depictions prompt prosocial behavior (Bateson et al., 2006; Haley & Fessler, 2005). Furthermore, when images of eyes are used in advertisements for green products, purchase intentions increase (Tong et al., 2020).

As such, eyes can be added to objects to make viewers perceive that the objects have natural human-like qualities (Haley & Fessler, 2005), emotions (Zhang et al., 2022) and even observational power (Bateson et al. 2006). Consequently, eyes are one of the commonly used cues for anthropomorphizing products (Kim & McGill, 2011; Kim et al., 2016). Aligned with the understandings that eyes are strong anthropomorphism cues, we designed our study so that participants assigned to the anthropomorphism condition viewed pictures of eggplants with googly eyes, while those in the non-anthropomorphism condition viewed plain eggplants without googly eyes (see Appendix).

During the study, participants imagined themselves in a grocery store and were informed that fresh produce was sourced from by *Del Campo*, a farm known for its special care of vegetables. After viewing the assigned ads at their own pace, participants rated the overall attractiveness of the eggplants using three items (*unattractive/attractive*, *ugly/beautiful*, and *inelegant/elegant*) on a seven-point semantic differential scale ($\alpha = 0.92$). Participants were then rated the extent to which the eggplants' appearance is ugly or pretty for produce shape manipulation checks and generic or personal for anthropomorphism manipulation checks on a seven-point semantic differential scale. Demographic measures were collected, and participants were debriefed and thanked upon completion.

3.3 | Results

The manipulation check showed that participants perceived the regular-appearing eggplants to be prettier than the irregular-appearing eggplants ($M_{regular}$ = 5.49 vs. $M_{irregular}$ = 3.77; t (165) = −6.62, p < 0.001). They perceived the eggplants with googly eyes to be more personal than the eggplants without googly eyes ($M_{googly\text{-}eyes}$ = 4.76 vs. $M_{no\text{-}eyes}$ = 3.88; t (165) = −3.42, p < 0.001).

To test the main hypothesis, we performed a two-way ANOVA on attractiveness. Produce shape had a significant main effect on attractiveness (F (1, 162) = 8.99, p < 0.01). However, anthropomorphism did not have a significant main effect (F (1, 162) = 0.41, p = 0.51). As H1 predicted, anthropomorphic eyes significantly interacted with produce shape (F (1, 162) = 5.77, p < 0.05).[1] As Figure 1 shows, contrasts indicated higher attractiveness for an irregular-appearing eggplant with (vs. without) googly eyes (Mgoogly-eyes = 4.00 vs. Mno-eyes = 3.17; t (79) = 2.25, p < 0.05). However, attractiveness was not significantly different for a regular-appearing eggplant with or without googly eyes (Mgoogly-eyes = 4.16 vs. Mno-eyes = 4.64; t (83) = −1.19, p = 0.23).

4 | DISCUSSION IN BRIEF

Study 1 provided initial insights into our hypotheses. We anticipated that incorporating anthropomorphic cues would enhance the perceived attractiveness of irregular-appearing produce. Supporting our expectations, participants rated misshapen eggplants as more attractive when adorned with googly eyes; however, this effect was not observed for regular-appearing produce. These findings are significant as they offer direct evidence of the role of attractiveness, demonstrating its applicability across broader populations while accounting for alternative explanations such as feelings of disgust and health concerns. Nevertheless, it is important to note certain limitations of the study. The downstream effects on consumer behavior were not assessed, and the findings are specific to a single type of fruit, namely eggplants, and a prominent visual cue of anthropomorphism, namely googly eyes, which will be addressed in the following studies.

[1]The primary aim of this paper is to determine if applying anthropomorphic cues to irregular-appearing produce could increase consumer purchase intentions by increasing the perceived attractiveness of the produce. To test this hypothesis, we compare produce with versus without anthropomorphic cues. In interpreting interactions, using or not using these anthropomorphic cues thus constitutes the between variable across our studies, and we examine this difference within the context of produce type—either regular-appearing or irregular-appearing.

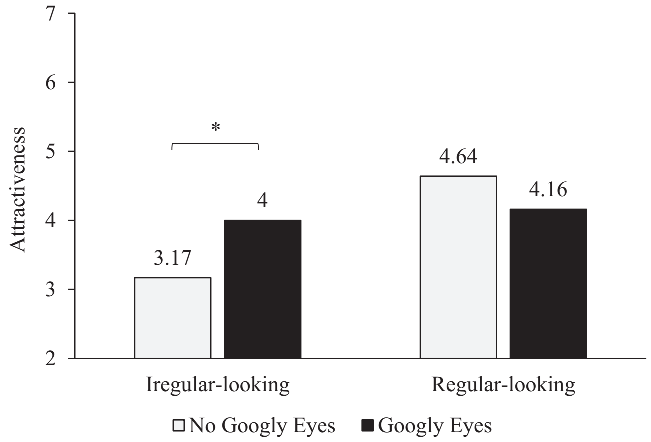

FIGURE 1 Two-way interaction between anthropomorphism and produce type on attractiveness (Study 1). * means the difference is significant at the 0.05 level.

4.1 | STUDY 2

The purposes of Study 2 were to examine the role of anthropomorphism in (1) consumption behavior of purchasing irregular-appearing produce and test this effect across a broader range of (2) produce types and (3) anthropomorphic cues. Study 2 anthropomorphized three different fruits, lemon, strawberry, and peach, through naming and hypothesized that study participants will express increased purchase intentions toward irregular-appearing produce (H2).

4.2 | Method

Study 2 had a 2 (produce shape: regular-appearing vs. irregular-appearing) × 2 (anthropomorphism: human-name vs. no-name) between-subjects design. Ninety-six college students from a North-eastern U.S. University participated (M_{age} = 19.86 years, 49% men). The procedure and measure of Study 2 mirror that of Study 1, except for three modifications.

First, we created two sets of ads for *Del Campo*. Each set of ads featured three fruits, such as lemon, strawberry, and peach, as spokes-characters. In one set, the irregular-appearing produce condition, the fruits had abnormal and nonstandardized appearances (e.g., a three-headed strawberry), while in the other set, the regular-appearing produce condition, the fruits had a symmetrical and balanced shape as a standard appearance (e.g., a one-head strawberry). Similar with Study 1, to isolate the effect of produce appearance, the ads message commonly highlights the health benefit of each fruit. For example, the ad message for Jordan the strawberry read, “Jordan (vs. the strawberry) offers a host of health benefits. It provides a hefty dose of vitamin C” (see Appendix).

Second, we manipulated anthropomorphisms by assigning human names to produces. Naming is another technique for anthropomorphizing inanimate entities. Indeed, in everyday life, more than 60% of individuals use naming to personify possessions and inanimate objects in various contexts (Brédart, 2021), such as cars or computers (Chartrand et al., 2008; Epley et al., 2007), information systems (Pfeuffer et al., 2019), intelligent personal assistants (Araujo, 2018; Cao et al., 2019; Laban, 2021; Seeger et al., 2021), and even product brands (Yang et al., 2020). Naming was shown to increase consumer trust toward autonomous vehicles (Waytz et al., 2014). Thus, naming can be a personal and commercial approach for personifying objects. Consistent with those findings, in Study 2, we gave common human names, such as "Jordan, the strawberry," as anthropomorphic cues for participants in the named condition. For participants in the no-name condition, the fruits were simply identified by type, such as "strawberry" (see Appendix).

Third, after viewing the assigned ads self-paced, participants indicated overall purchase intentions for the fruits from the *Del Campo* farm with a four-item (*unlikely/likely*, *would not/would*, *impossible/possible*, *improbable/probable*) on a seven-point semantic differential scale ($\alpha = 0.94$). Attractiveness was not measured. For the manipulation checks, participants rated the fruit featured in the ad as either ugly or pretty and either generic or personal on a seven-point semantic differential scale. After completing demographic measures, participants were debriefed and thanked.

4.3 | Results

The manipulation check showed that participants perceived the irregular-appearing produces to be uglier than the regularly shaped produces ($M_{regular} = 5.78$ vs. $M_{irregular} = 3.94$; t (94) = −5.83, $p < 0.001$). They perceived the named fruits to be more personal than the unnamed fruits ($M_{name} = 5.10$ vs. $M_{no\text{-}name} = 4.51$; t (94) = 1.61, $p < 0.05$).

To test the main hypothesis, a two-way ANOVA was performed on purchase intention. Produce shape had a significant main effect on purchase intention (F (1, 92) = 6.42, $p < 0.05$), indicating higher purchase intentions for regular-appearing fruits ($M_{regular} = 5.18$ vs. $M_{irregular} = 4.42$; t(94) = −5.83, $p < 0.001$). However, anthropomorphism did not have a significant main effect (F (1, 92) =0.746, $p = 0.39$). As H1 predicted, an interaction between the anthropomorphic naming and produce shape was significant (F (1, 92) = 6.68, $p < 0.05$). As Figure 2 shows, contrasts indicated higher purchase intentions for irregular fruits with (vs. without) human names (Mname = 4.98 vs. Mno-name = 3.90; t (48) = 2.46, $p < 0.05$). However, purchase intentions were not significantly different for regular fruits with or without human names (Mname = 4.97 vs. Mno-name = 5.50; t (44) = −1.21, $p = 0.23$).

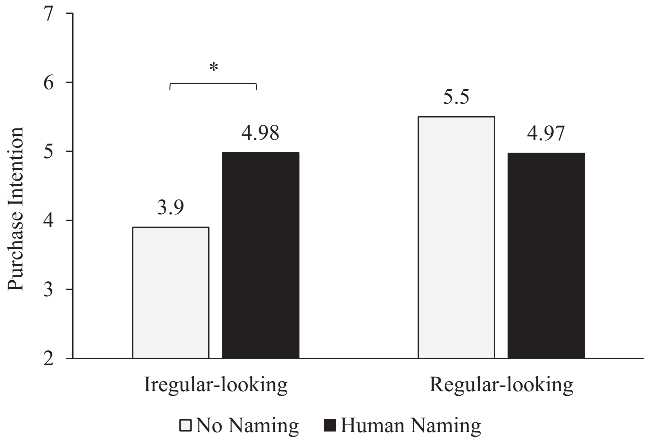

FIGURE 2 Two-way interaction between anthropomorphism and produce type on purchase intention (Study 2). * means the difference is significant at the 0.05 level.

5 | DISCUSSION IN BRIEF

Study 2 provides evidence that anthropomorphizing increases purchase intentions for irregular produce only. When human names such as Jordan, Alex, or Taylor are assigned to irregular-appearing fruits, such as a three-headed strawberry or four-tailed lemon, participants indicate being more likely and willing to purchase them. However, this increased preference is not observed in ads featuring fruits in standard shapes. The finding supports our contention that the positive effect of anthropomorphism does not merely arise from the imputation of human attributes to produce. Rather, by providing human-like names to irregular produce, consumers may consider a variety of esthetic cues, potentially reducing the negative impact of irregular-appearing appearance and increasing purchase intentions. Although the results support our predictions, the positive effects of anthropomorphism on consumer preference for irregular-appearing produces may not be universally applicable. Moreover, it remains unclear whether these effects can be generalized to other produce categories, such as vegetables. Furthermore, Study 1 and Study 2 evaluated attractiveness and purchase intentions separately, without directly examining whether heightened attractiveness mediates the positive impact of anthropomorphism on consumer purchase intentions. To address these issues, we conducted Study 3.

5.1 | STUDY 3

The aims of Study 3 were: (1) to replicate the findings of Study 1 and Study 2 using vegetables as the primary produce category; (2) to extend our understanding by including two outcome variables, attractiveness and purchase intention; (3) to investigate the moderating role of farm type (i.e., corporate vs. local), proposing that the naming effect holds for produce from corporate farms but diminishes for produce originating from local farms (H3); and (4) to examine the mediation hypothesis (H3), aiming to identify attractiveness as the underlying mechanism driving increased purchase intentions toward anthropomorphized produces.

5.2 | Method

Study 3 had a 2 (produce shape: regular-appearing vs. irregular-appearing) × 2 (anthropomorphism: human-name vs. no-name) × 2 (farm type: corporate vs. local) between-subjects design. Participants were 208 undergraduate students from a northeastern U.S. university (M_{age} = 19.82 years, 60% men).

The procedure and measures were the same as those in Study 2, except for the following modifications. First, we manipulated farm type. Participants in the local farm condition, *Del Campo* was depicted as "a small local family farm in southern New England, offering the best local selection through hand-sorting." Participants viewed a video clip showing human hands placing tomatoes in boxes. For participants in the large corporate farm condition, *Del Campo* was depicted as "an agricultural corporation that offers the best industry selection" and uses high-tech engineering to facilitate nationwide distribution. Participants viewed a video clip showing robotic hands boxing tomatoes (see Appendix).

We conducted a separate pretest to confirm whether we manipulated farm type as intended. Ninety-six undergraduate students from a northeastern U.S. university were randomly assigned to the corporate or local farm condition. After participants viewed the stimulus ads and information about the farm, they answered questions that assessed their perceptions. Specifically, they evaluated whether the farm followed traditional or industrialized practices. Using a series of sliding scales, they estimated farm size in terms of acreage (1–10,000 acres), number of employees (1–1000 employees), pounds of produce per acre (1–100,000 pounds), and annual revenue in U.S. dollars ($1000–$100 million). Results revealed that participants in the corporate farm (local farm) condition saw the farm as more industrial (traditional) (M_{local} = 2.79 vs. $M_{corporate}$ = 5.77; t (93) = 10.58, $p < 0.001$). They estimated the corporate (local) farm to be larger (smaller) in terms of acreage (M_{local} = 3,208 vs. $M_{corporate}$ = 4580 acres; t (92) = 2.66, $p < 0.01$), number of employees (M_{local} = 256 vs. $M_{corporate}$ = 372 employees; t (92) = 3.59, $p < 0.01$), pounds of produce per acre (M_{local} = 27,913 vs. $M_{corporate}$ = 46,013 pounds per acre; t (92) = 3.59, $p < 0.001$), and annual revenue in U.S. dollars (M_{local} = 26 million vs. $M_{corporate}$ = 46 million U.S. dollars; t (92) = 3.95, $p < 0.001$).

Second, we created two sets of advertisements promoting three vegetables: eggplants, tomatoes, and carrots. Each set presented the same vegetables, albeit with either regular- or irregular-appearing and with or without human names (e.g., Taylor, the eggplant; Alex, the tomato; Jordan, the carrot). Third, participants measured the attractiveness of the assigned produce and indicated their willingness to purchase the items using the same items used in Study 1 and Study 2. Additionally, for the farm type manipulation checks, participant rated whether the vegetable was cultivated and produced by a corporate farm or a local farm on a seven-point scale.

5.3 | Results

The manipulation check showed that participants perceived the regular-appearing vegetables to be prettier than the irregular-appearing vegetables ($M_{regular}$ = 4.99 vs. $M_{irregular}$ = 3.27; t (206) = −7.58, $p <$ 0.001). Participants also perceived the human-named vegetables to be more personal than the unnamed vegetables (M_{name} = 4.84 vs. $M_{no\text{-}name}$ = 4.21; t (206) = 2.20, $p < 0.05$). Additionally, they perceived the robotic sorting used in the corporate farm condition to be more characteristic of a corporate farm than the human sorting used in the local farm condition ($M_{corporation}$ = 4.65 vs. M_{local} = 3.00; t (206) = −6.00, $p < 0.001$).

To test the main hypothesis, we performed a three-way MANOVA on attractiveness and purchase intention. As for attractiveness perceptions, produce shape had a significant main effect (F (1, 200) = 16.45, $p < 0.001$). However, anthropomorphism (F (1, 200) = 0.38, p = 0.53) and farm type (F (1, 200) = 0.49, p = 0.48) had no significant main effects. As H3 predicted, a significant three-way interaction was found for attractiveness (F (1, 200) = 8.37, $p < 0.01$). Specifically, for the corporate farm, anthropomorphic naming had a significant two-way interaction with produce shape (F (1, 92) = 5.07, $p < 0.05$). For the local farm, however, the interaction was non-significant (F (1, 108) = 3.45, p = 0.06). As Figure 3 shows, for corporate farm produce, contrasts revealed that participants reported more (less) attractiveness for irregular vegetables with (without) anthropomorphic names (Mname = 4.33 vs. Mno-name = 3.31; t (48) = −2.32, $p < 0.05$). For regular vegetables however, attractiveness showed no significant interaction (Mname = 4.21 vs. Mno-name = 4.65; t (44) = 0.92, p = 0.36). The interaction effect disappeared for the local farm, but vegetable shape had a significant main effect: regular (irregular) vegetables were perceived to be more (less) attractive (Mregular = 4.63 vs. Mirregular = 3.34; F (1, 108) = 15.27, $p < 0.001$).

Purchase intention also had the same results as attractiveness. Produce shape had a significant main effect on purchase intention (F (1, 200) = 13.39, $p < 0.001$), however, anthropomorphism (F (1, 200) = 0.06, p = 0.80) and farm type (F (1, 200) = 0.27, p = 0.60) did not have significant main effects. As H3 predicted, a significant three-way interaction was found for purchase intention (F (1, 200) = 4.91, $p < 0.05$). Specifically, for the corporate farm, a two-way interaction between the anthropomorphic naming and produce shape was significant (F (1, 92) = 8.31, $p < 0.01$). For the local farm, however, the interaction was nonsignificant (F (1, 108) = 0.03, p = 0.86). As Figure 4 shows, for corporate farm produce, contrasts revealed that participants reported higher purchase intentions for irregular vegetables with anthropomorphic names compared to irregular vegetables without anthropomorphic names (Mname = 4.53 vs. Mno-name = 3.42; t (48) = −2.19, $p < 0.05$). However, purchase intentions for regular vegetables showed no significant interaction (Mname = 4.18 vs. Mno-name = 4.97; t (44) = 1.68, p = *n.s.*). For the local farm, the interaction effect disappeared, but vegetable shape had a significant main effect: participants indicating higher buying intentions for regular vegetables than for irregular vegetables (Mregular = 4.92 vs. Mirregular = 3.87; F (1, 108) = 11.58, $p < 0.01$).

We hypothesize that heightened attractiveness explains higher purchase intention for anthropomorphized produce from a corporate farm, but not from a local farm. To further analyze H3, we conducted

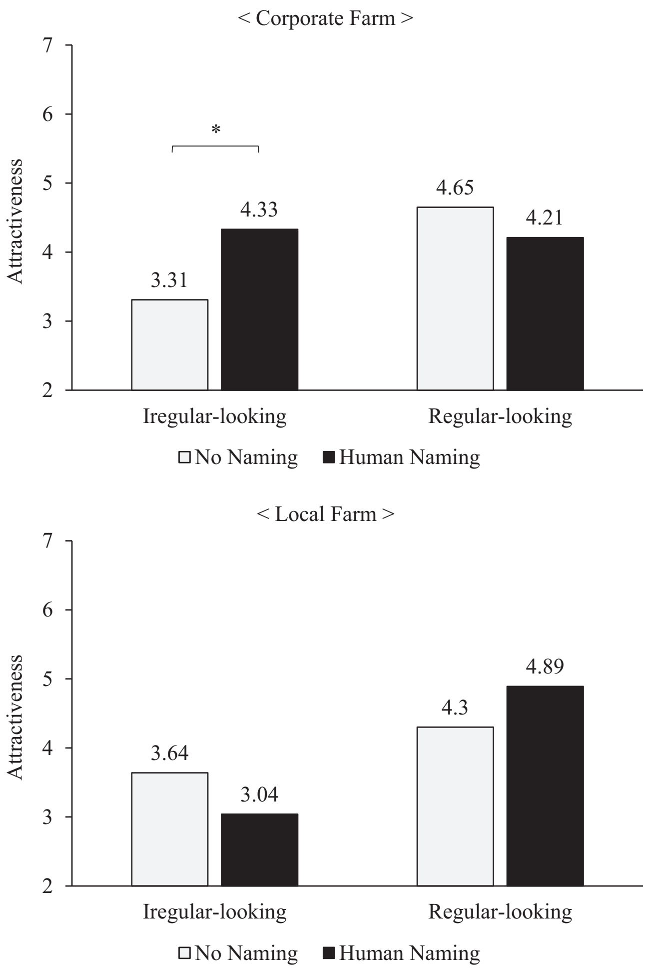

FIGURE 3 Three-way interaction between anthropomorphic naming, fruit types, and farm type on attractiveness (Study 3). *means the difference is significant at the 0.05 level.

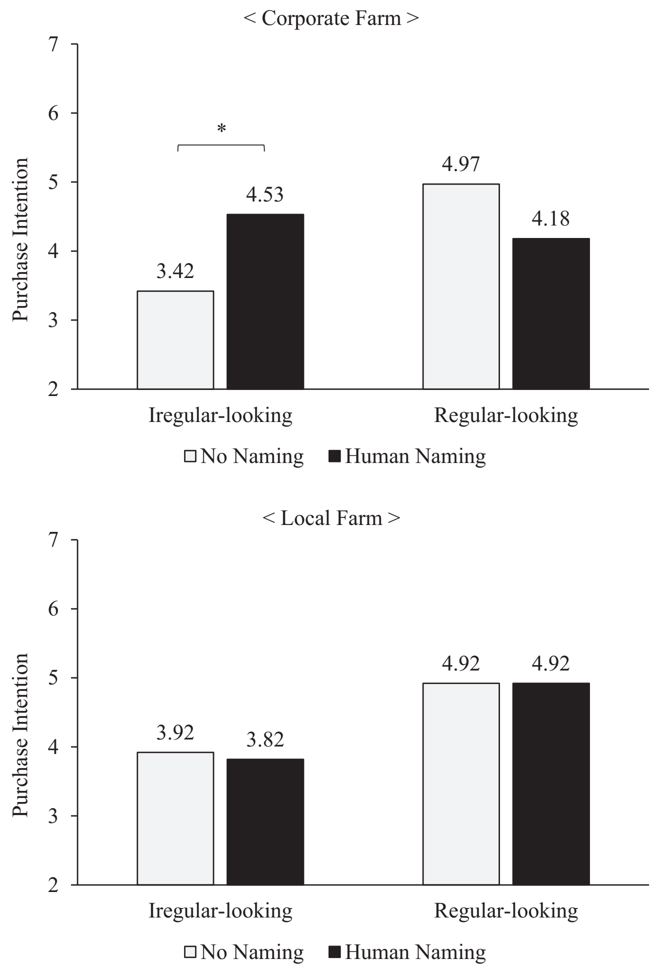

FIGURE 4 Three-way interaction between anthropomorphic naming, fruit types, and farm type on purchase intention (Study 3). *means the difference is significant at the 0.05 level.

a moderated mediation analysis performed separately for a corporate and local farm condition (Hayes 2018; Model 7, 5000 bootstrap samples) where anthropomorphism as an independent variable (X), produce shape as a moderating variable (W), attractiveness as a mediating variable (M), and purchase intention as a respective dependent variable (Y).

As shown in Figure 5, for the corporate farm, the results yielded a significant moderated mediation effect (Index = −1.05, SE = 0.5, CI = −2.1 to −0.12). Specifically, there was a significant indirect effect (IE) of anthropomorphism on purchase intentions through perceptions of attractiveness only for irregular-appearing produce ($b = 0.74$, $SE = 0.34$, CI = 0.11–1.45). This suggests that when the produce is irregular-appearing, names (vs. no-names) increases the attractiveness of the produce ($b = 1.03$, $SE = 0.45$, CI = 0.14–1.91). The increased attractiveness then increases purchase intention ($b = 0.72$, $SE = 0.09$, CI = 0.53–0.9). However, when produce is regular-appearing, the indirect effects of anthropomorphism on purchase intention through attractiveness were insignificant ($b = -0.32$, $SE = 0.36$, CI = −1.03 to 0.38). For the local farm, the moderated mediation analysis yields insignificant effects for purchase intention through attractiveness (Index = 0.59, $SE = 0.32$, CI = −0.03 to 1.25), suggesting that anthropomorphisms did not impact attractiveness and purchase intention for both irregular-appearing ($b = -0.29$, $SE = 0.25$, CI = −0.79–0.21) and regular-appearing produce ($b = 0.29$, $SE = 0.2$, CI = −0.08 to 0.71). Thus, H3 was supported.

6 | DISCUSSION IN BRIEF

Study 3 not only established the boundary conditions under which anthropomorphism promotes the consumption of irregular-appearing produce but also provided further support for our exemplar-based account of consumers' reluctance to purchase irregular-appearing food and the remedial effect of anthropomorphism. Specifically, the moderating role of farm type offers indirect evidence that consumers tend to reject irregular-appearing produce because they perceive the irregular appearance as a violation of the exemplar rule. However, as evident in the significant impact of anthropomorphism and the mediating role of attractiveness, we demonstrated that this detrimental effect can be mitigated if consumers actively process multiple

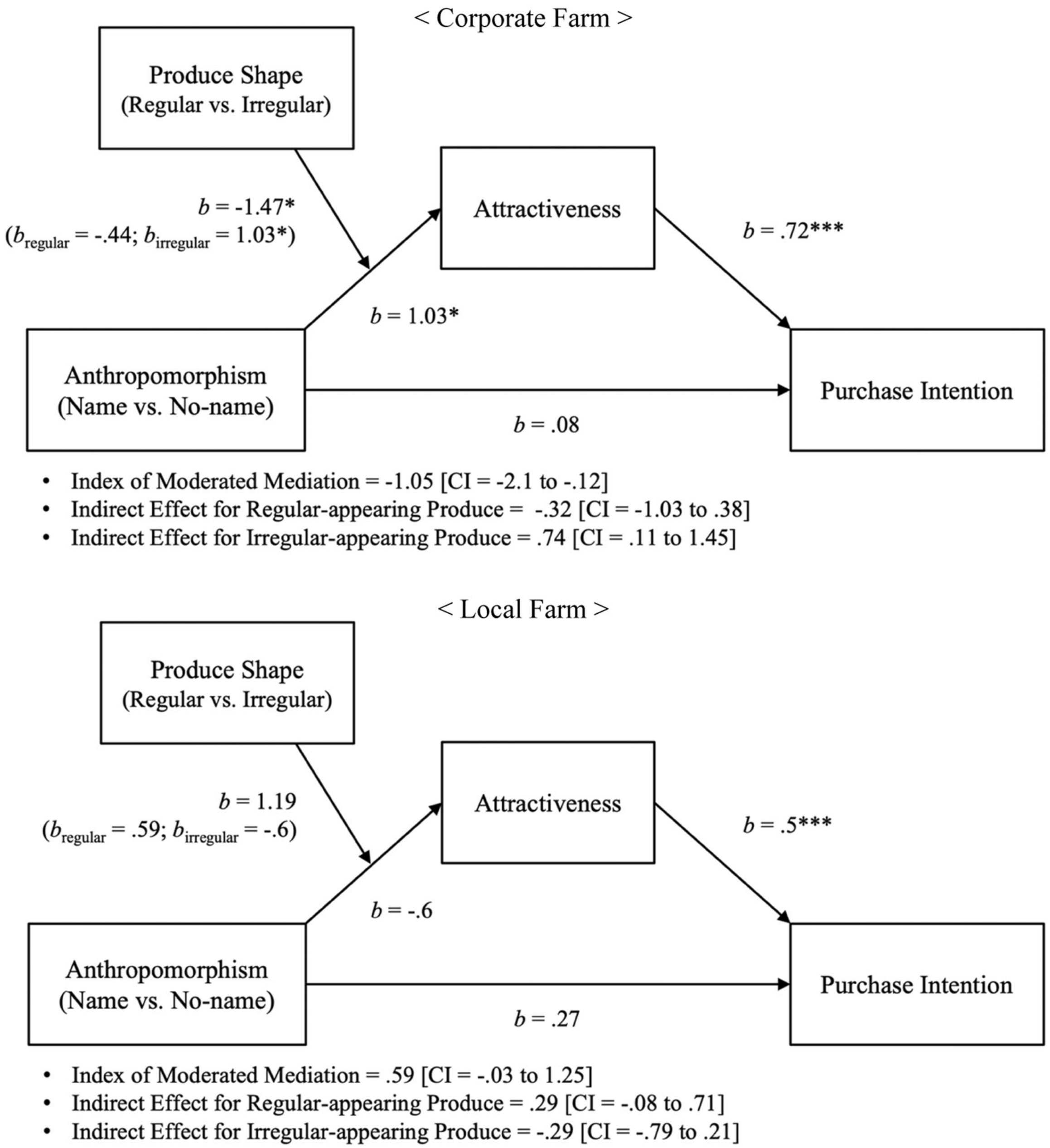

FIGURE 5 Moderated mediation (Study 3). *<0.05, **<0.01, ***<0.001.

exemplars. Importantly, we obtained supportive results from three different vegetables: eggplants, tomatoes, and carrots. The consistent findings across diverse produce types further corroborate the robustness of our theoretical framework.

7 | GENERAL DISCUSSION

7.1 | Theoretical implications

Reducing food waste is essential as the population continues growing and natural resources are being exhausted. Our study aims to investigate innovative strategies for encouraging consumers to purchase irregular-appearing produce that would otherwise be wasted. A key theoretical contribution of our research lies in leveraging exemplar theory (Kahneman & Miller, 1986) to explain consumers' hesitancy toward buying such produce. We also propose the use of anthropomorphism by retailers to alter consumer perceptions and drive purchase decisions. This significantly enriches our understanding of exemplar theory in multiple dimensions.

First, exemplar theory, originated from cognitive psychology, has primarily been explored within the interpersonal domain, focusing on various cognitive process related to memory (Hugenberg & Sacco, 2008; Smith, 1988), social judgment (Smith & Zárate, 1992), categorization (Lacroix et al., 2005; Smith & Zarate, 1990), and stereotyping (Duval et al., 2018; Lee et al., 2024; Windels et al., 2024).

Only a few studies have examined the potential of exemplar theory in consumer psychology (Loken et al., 2008). Addressing this gap, this paper ventures into new territory by expanding the application of exemplar theory to explain consumers' esthetic reactions to produce. Specifically, we discover that consumer rejection to ugly produce is mainly due to the deviations in its shape from typical exemplar or norm, subsequently influencing judgments of attractiveness and desirability (Zhang & Rodgers, 2023). Additionally, researchers have pointed out the limited focus on the practical application of this theory in addressing socio-environmentally issues (Georgeac & Rattan, 2019; Hegarty, 2017). To address this gap, this paper highlights the relevance and applicability of exemplar theory in tackling real-world challenges as a problem-solving tool, particularly concerning consumer reluctance towards irregular-appearing produce, where conventional strategies may fall short.

Second, this paper introduces a novel utilization of exemplar theory through anthropomorphism. Although prior research has centered on the concept of "counter-exemplars" to address the negative impact of violating standard exemplars—presenting opposing examples to challenge stereotypes—(Ramasubramanian, 2011; Stapel & Koomen, 1998), our approach diverges. Instead of directly countering the exemplar with its opposite, we introduce anthropomorphism as a method to concurrently consider multiple exemplars, reducing reliance on a singular exemplar. This departure from traditional methods is significant as it offers a more nuanced and flexible approach to utilizing exemplar theory. By embracing anthropomorphism, we encourage individuals to view products with diverse qualities and characteristics, akin to how they perceive different traits in humans. This approach not only addresses the rigidity associated with relying on a single exemplar but also allows for broader and more inclusive judgments.

Lastly, we demonstrate that exemplar theory can offer a nuanced understanding of esthetic cues and personal experiences. While other theories, such as schema congruency (Taylor & Crocker, 2022), appraisal (Briñol et al., 2018), goal conflict (Stroebe et al., 2017), or stereotype theories (Kervyn et al., 2012), may provide valuable insights, they often offer only "partial" explanations for our findings. For instance, while congruence theory (Taylor & Crocker, 2022) may offer insights into consumers' reluctance towards irregular-appearing produce and the influence of farm types, the intricacies of anthropomorphism and its moderating role in consumer resistance may surpass the straightforward notions of congruence and incongruence in produce appearance. As a result, the mere binary categorization of congruence might not fully elucidate the complexities of three-way interactions, and more nuanced dynamics could be at play within consumers' cognitive processes. Appraisal theory (Briñol et al., 2018) focuses on how consumers initially evaluate potential risks or quality. The theory has limited applicability in our context because we constantly expose study participants to health-related information across conditions, eliminating health concerns that might elicit disgust or aversion. Goal conflict theory (Stroebe et al., 2017) might explain why the reality of corporate produce disappoints consumer expectations, but our studies set no specific consumer goals, so the theory fails to address how anthropomorphism influences aversion. Stereotype theories like the BIA framework (Kervyn et al., 2012) may explain perceptions that brands have warmth and competence, but they do not explain complex cognitive responses to product attributes like shape.

In contrast, exemplar theory (Kahneman & Miller, 1986) explains the complex cognitive processes of why consumers are repelled by irregular-appearing produce and why anthropomorphism can mitigate their aversion. Furthermore, the differing expectations of consumers regarding the production of regular-appearing produce in corporate versus local farms create conditions wherein anthropomorphism plays a crucial role in diversifying exemplars of produce appearance. Nevertheless, it is important to note that we do not view alternative theories as competing with exemplar theory; rather, we see them as complementing each other, providing rich explanations in their own ways (Loken et al., 2008). For instance, individuals must have established exemplars to determine consistency between concepts, evoke appraisal, gauge goal conflict, or identify stereotypes. When individuals lack experience or memories about an object or event, they may struggle to form strong exemplars, disrupting their determination of congruency, appraisal, goal conflicts, or stereotypes. Thus, in grasping the relationship between these theories, we believe that exemplar theory can serve as a fundamental framework for encompassing the respective effects of alternative theories.

8 | PRACTICAL IMPLICATIONS

Our research provides practical implications for the food industry, as it suggests that anthropomorphizing strategies can effectively increase demand for irregular-looking produce. Because most imperfect produce is wasted before it reaches grocery stores, we suggest that marketing communicators can subtly shift presentations about produce, at or before the point of sale, as a crucial first step toward conserving resources. By altering cosmetic standards and preemptively enhancing perceived attractiveness, we can boost consumer demand for irregular produce in the early stages of consumption, effectively changing negative perceptions about its appearance.

Anthropomorphism is a common marketing tactic for misshapen produce globally. Intermarché in France features "The Grotesque Apple" in its "Inglorious Fruits and Vegetables" campaign, adding humor to product personification (SAP BrandVoice, Forbes, 2019). In the UK, Asda's "Beautiful on the Inside" line emphasizes that flavor and nutrition outweigh looks (The Guardian, 2015). Loblaws in Canada uses the "Naturally Imperfect" label to showcase the unique traits of its produce, likening them to human personalities (Liu, 2016). In the US, Giant Eagle's "Produce with Personality" campaign in Pittsburgh also employs this strategy (The Produce News, 2018). Retailers can leverage these findings by implementing such strategies and emphasizing the unique and valuable characteristics of irregular-looking produce. For example, grocery stores can give human names to irregular-looking produce and place them in prominent areas with appealing signage, creating a connection between consumers and the produce. This can also be coupled with informational campaigns highlighting the nutritional benefits of irregular-looking produce and how it contributes to a sustainable food system.

Our findings have straightforward implications for large national grocery chains. We found strong support for the effectiveness of anthropomorphism in promoting the consumption of irregular-looking produce, countering the negative impact of its irregular appearance on consumers' purchase intentions. Moreover, the effectiveness of this strategy is contingent on farm type, with consumers being more receptive to anthropomorphism when produce is sourced from a large corporate farm, as opposed to a small local farm (Jeong, 2024). When our study participants viewed a video clip of produce being sorted and boxed by robotic hands in a large corporate farm, they were more receptive to the positive effects of anthropomorphism. They were more resistant, however, when they viewed a clip of produce being sorted and boxed by human hands at a small local farm. This underscores the need for corporate retailers to play a pivotal role in promoting the consumption of irregular-looking produce through anthropomorphizing strategies.

Emphasizing the role of irregular produce in achieving food security and waste reduction can integrate into national policies, underscoring its importance in a sustainable food ecosystem. A key area for policy-driven change is the K-12 education system, where introducing educational initiatives that highlight the value of all produce, regardless of appearance, can foster an early appreciation for food diversity and sustainability among students. This could include developing curriculum modules focused on the importance of minimizing food waste, the nutritional benefits of consuming all types of produce, and the environmental advantages of such practices. Developing curriculum modules focused on minimizing food waste, the nutritional benefits of consuming all types of produce, and the environmental advantages of such practices can complement these educational efforts. Additionally, organizing field trips to local farms or farmers' markets allows direct engagement with farmers, who can discuss food waste challenges and the quality of irregular produce. Such interactions can dispel misconceptions and make learning about "ugly" fruits and vegetables engaging and fun through competitions and creative storytelling.

By facilitating partnerships between schools, local farms, and retailers, policymakers can ensure that educational efforts are complemented by accessible and available irregular produce, reinforcing the message that all food has value. This comprehensive approach enables policymakers not only to educate the younger generation about the importance of embracing all produce but also to instigate a cultural ship that influences broader community and family attitudes, paving the way for lasting changes in consumer behavior.

Moreover, leveraging social media platforms can extend the research of these messages, making the concept of irregular-looking produce more appealing to a younger, digitally savvy audience (Buckley et al., 2024; Haikel-Elsabeh, 2023; Saad, 2023; Wan & Jiang, 2023). Anthropomorphism and diversity messages can be used to make irregular-looking produce more appealing and to encourage consumers to proactively seek unique and differently shaped produce. For example, posting pictures of irregular-looking produce with quirky captions and giving them human-like names can make them more relatable and attractive to the public.

In summary, our research advocates for a concerted effort among educators, policymakers, and retailers to change perceptions and practices regarding the consumption of irregular-looking produce. Such efforts can lead to a more sustainable food system, reduce waste, and instill values of environmental conservation and food ethics in the next generation.

9 | LIMITATIONS AND FUTURE RESEARCH

Although grocery retailers can derive practical benefits from learning that they can use anthropomorphism to sell produce that might have been wasted, future research should address the limitations of our studies. First, we focus on how anthropomorphism increases purchase intentions and willingness to buy, without measuring actual consumption. Thus, future studies should capture actual consumption patterns to verify whether positive intentions translate to real-world purchasing and consumption behaviors. Additionally, future research should explore whether using anthropomorphizing to promote consumption has long-term sustainability implications for overall food systems or has possible unintended consequences on demands for other types of produce.

Second, future research might further examine how cognitive expectations interact with emotional reactions. Food products have unique physical and emotional health impacts. The "law of sympathetic magic in disgust" concept explains that our natural aversions to certain types of foods, often manifesting as disgust, protect us against potential toxins or harmful substances (Rozin et al., 1986). Although we present identical health benefits across our studies, future research may investigate how disgust shapes food preferences.

Third, exemplar theory indicates that new experiences cause people to dynamically adapt and evolve (Kahneman & Miller, 1986). That is, consumers may initially categorize irregular-appearing produce as atypical because it clashes with their stored exemplars, but they might form a new, more accepting category after they are exposed repeatedly and realize that fresh produce tastes the same, whatever its appearance. For instance, the Saturn peach used in our research can be seen as a new produce category and is not perceived as misshapen despite its deviation from regular-looking peaches. Thus, future exploration could consider how past experience and knowledge impact attitudes toward irregular-appearing produce. Moreover, consumers who enjoy novelty and unique experiences might perceive irregular-appearing produce as intriguing and distinctive, even without anthropomorphic cues. Therefore, future research could assess individual difference variables, such as the need for uniqueness (Tian et al., 2001).

In our studies, "ugly" refers to produce that deviate from standard expectations in terms of shape and appearance. Yet, our perception of produce ugliness should extend to other sensory attributes. For instance, consider the distinctive pale-yellow durian fruit, which is covered with a tough, spiky exterior and emits a notoriously strong odor that many find unpleasant. Those prominent features could set a unique standard for assessing produce ugliness, aside from its within-category typicality, suggesting that different perceptions of ugliness may determine whether anthropomorphism is effective.

Lastly, an intriguing avenue for future investigation is whether naming irregular-appearing produce alters flavor perceptions. Some evidence suggests that when humans interact with products, they affect product performance. For instance, when experimenters exposed plants to classical or jazz music, the plants grew more vigorously than plants left alone (Retallack, 1973). Similarly, when cows were named, they produced more milk than unnamed cows (Bertenshaw & Rowlinson, 2009).

In conclusion, our research provides novel insights into how exemplar theory and anthropomorphism can be employed to tackle the pressing issue of food waste. Furthermore, we offer actionable recommendations to reduce food waste and promote a more sustainable and ethical food system. By leveraging the power of anthropomorphism and expanding consumers' mental representations of produce categories, we aim to encourage a food system that values the unique and diverse qualities of all members of the category, including irregular-appearing produces.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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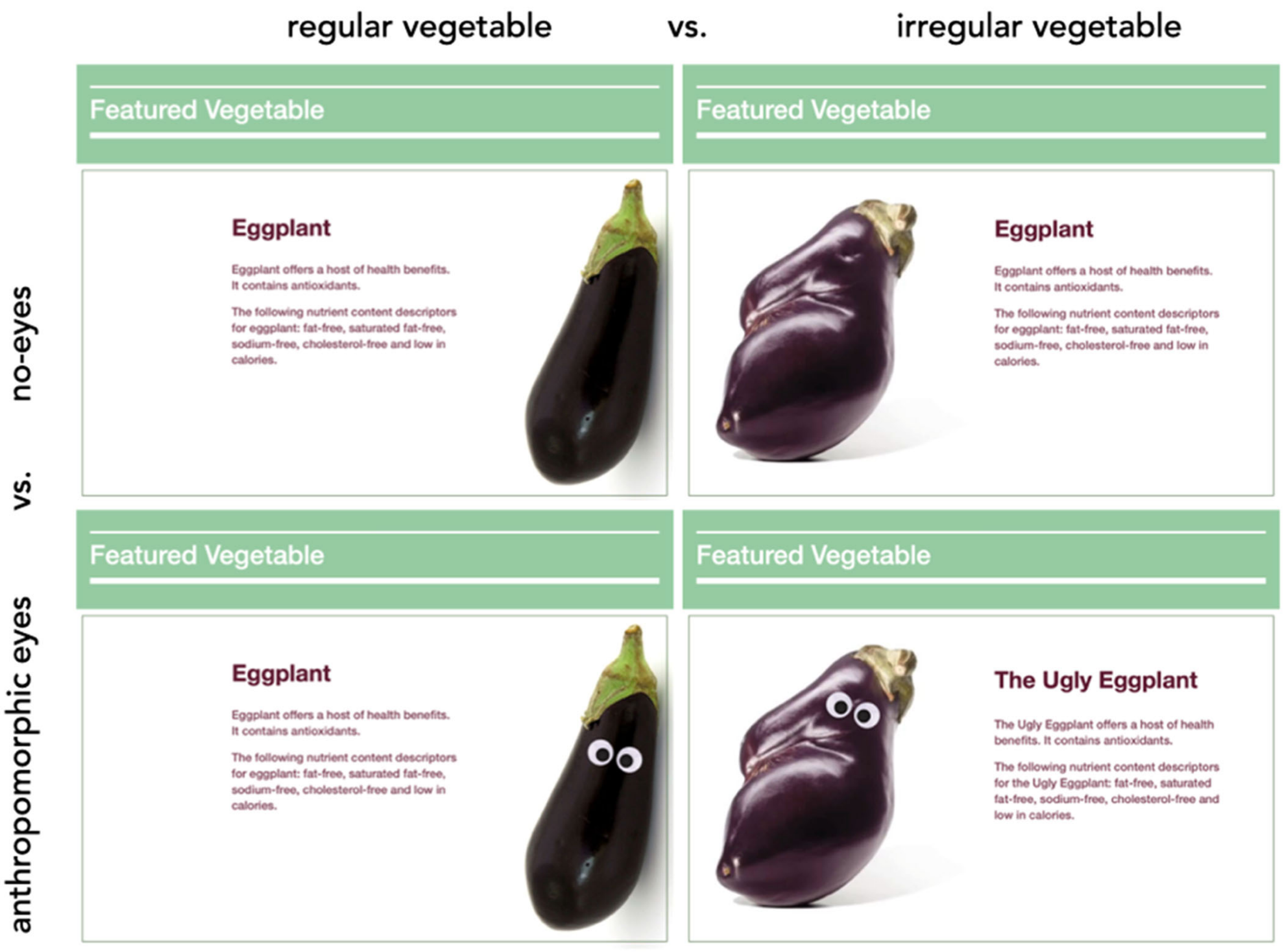
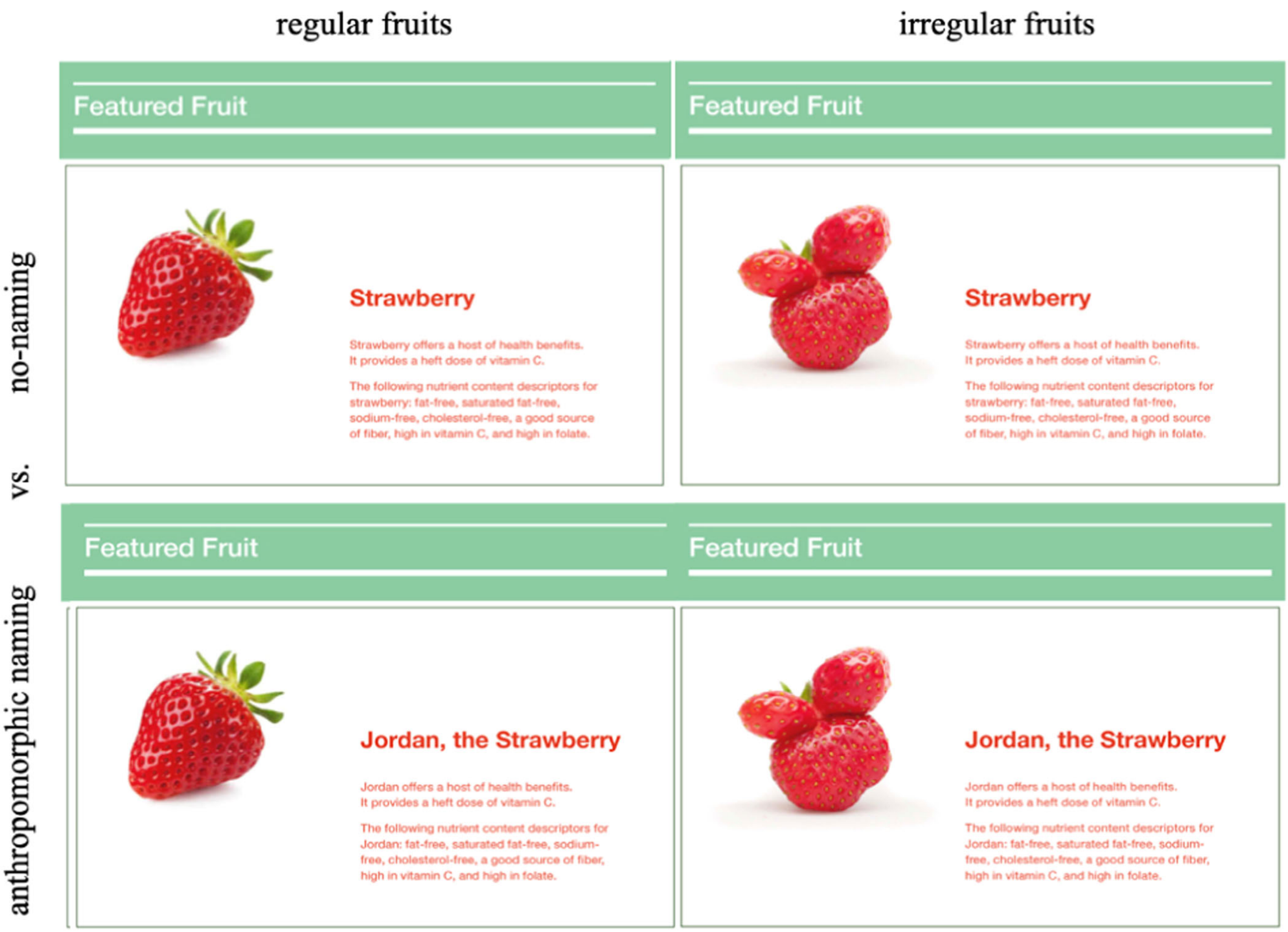
How to cite this article: Kim, K., Ryoo, Y., Manika, D., Yoon, N., & Yoon, S. (2024). From ugly to attractive: Leveraging anthropomorphism to increase demand for irregular-appearing produce. *Psychology & Marketing*, *41*, 2033–2056. https://doi.org/10.1002/mar.22025

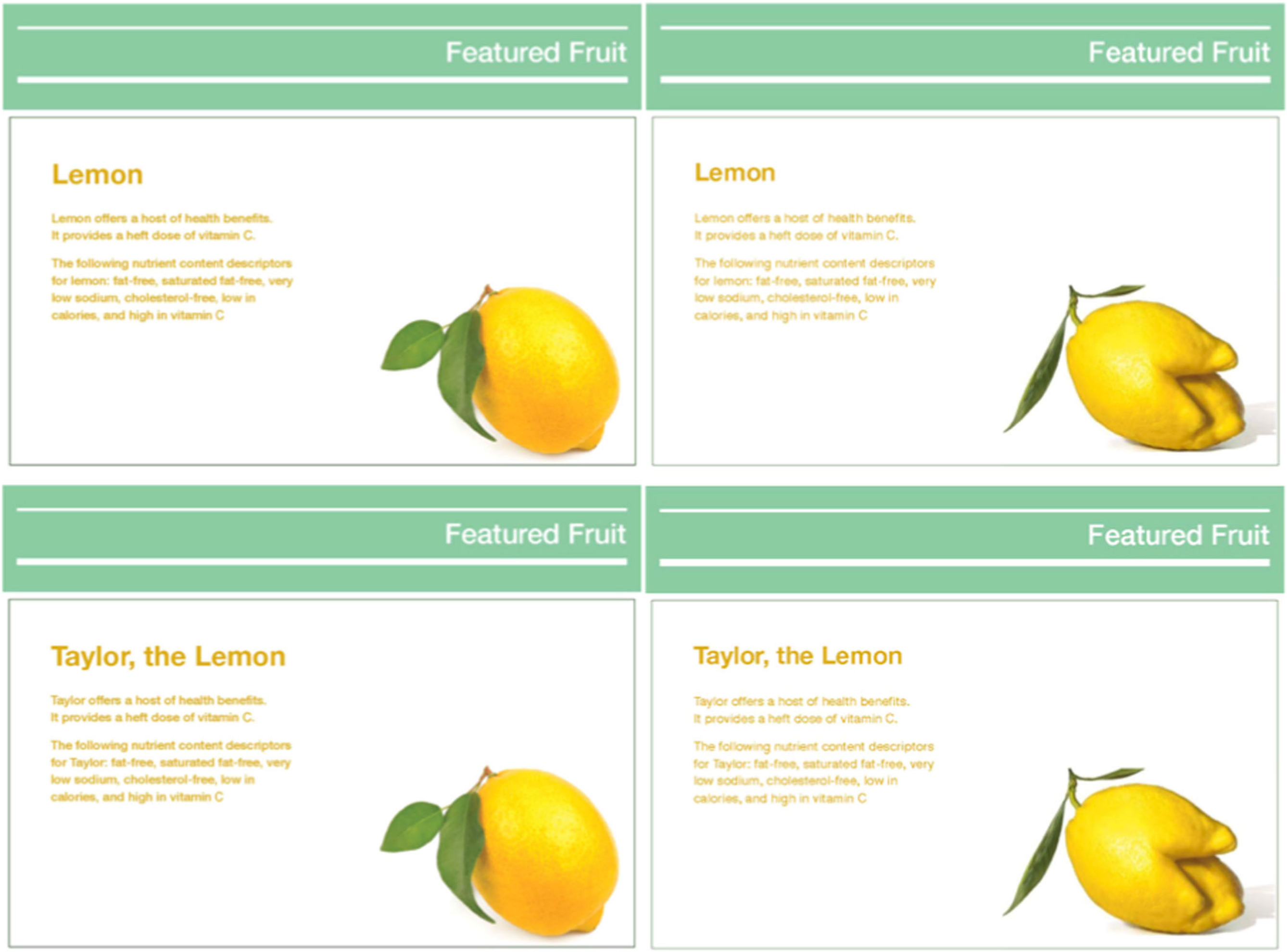
APPENDIX

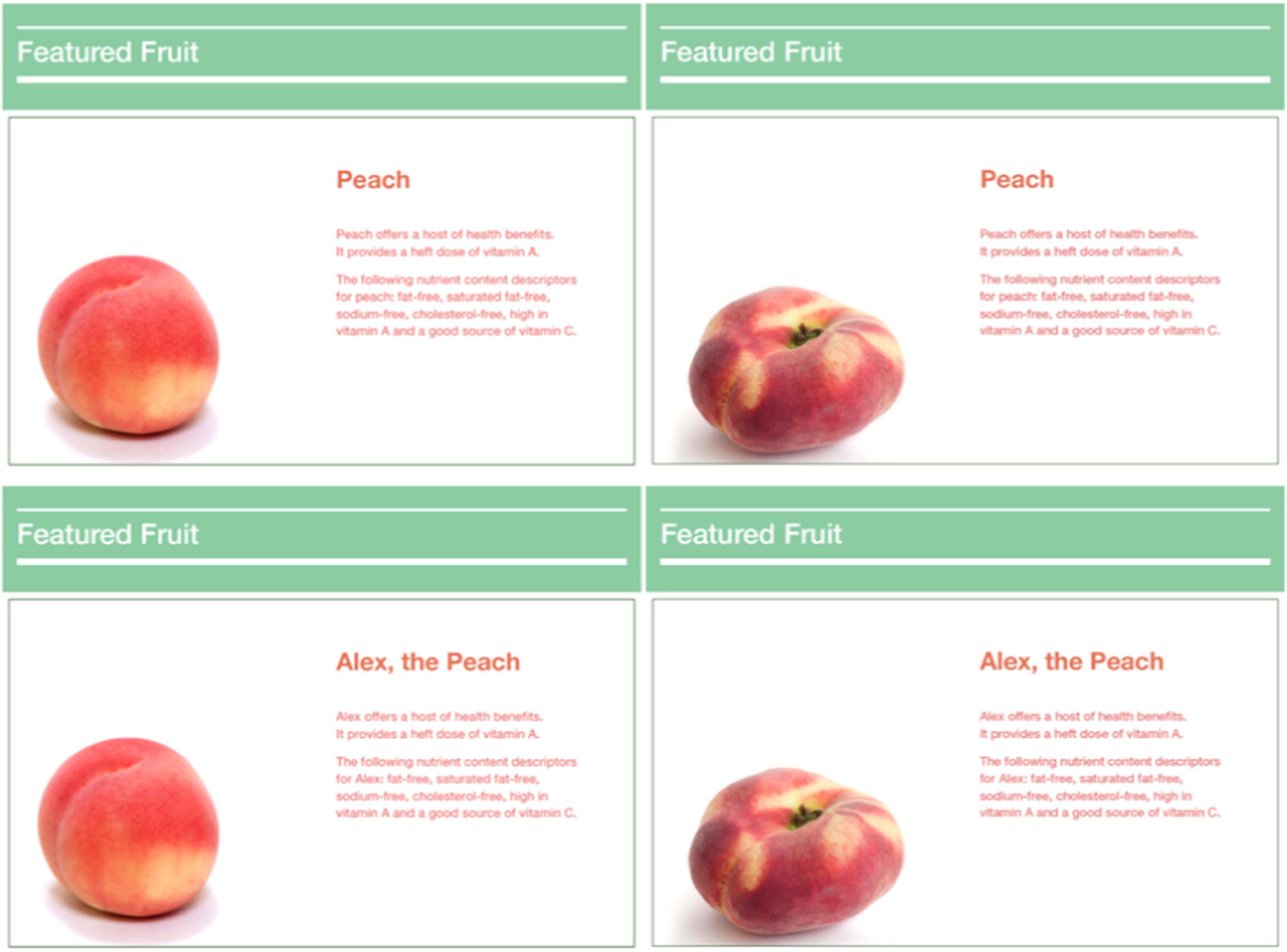
Study 1 Stimuli



Study 2 Stimuli



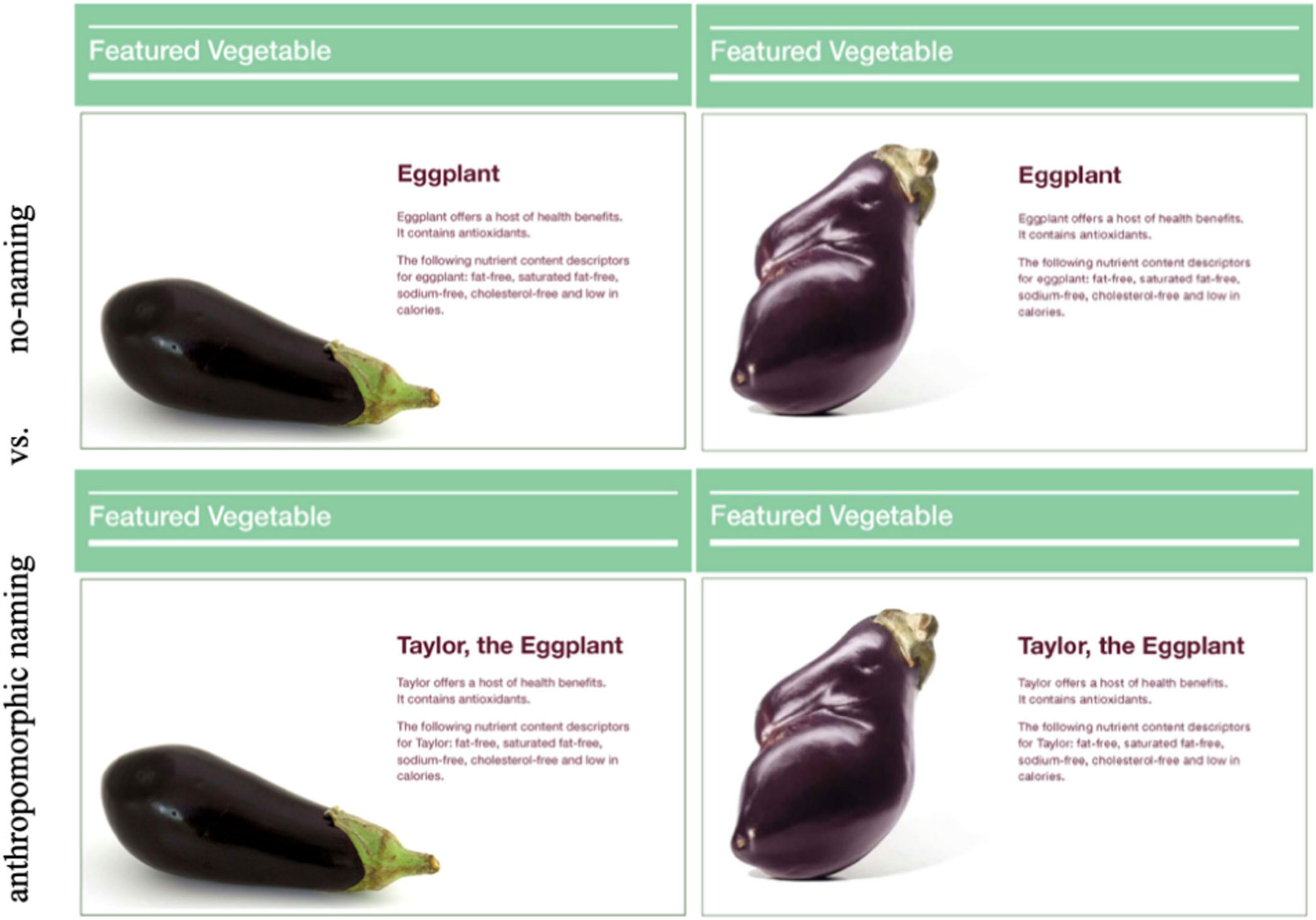

Featured Fruit
Lemon
Lemon offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for lemon: fat-free, saturated fat-free, very low sodium, cholesterol-free, low in calories, and high in vitamin C
Featured Fruit
Lemon
Lemon offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for lemon: fat-free, saturated fat-free, very low sodium, cholesterol-free, low in calories, and high in vitamin C
Featured Fruit
Taylor, the Lemon
Taylor offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for Taylor: fat-free, saturated fat-free, very low sodium, cholesterol-free, low in calories, and high in vitamin C
Featured Fruit
Taylor, the Lemon
Taylor offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for Taylor: fat-free, saturated fat-free, very low sodium, cholesterol-free, low in calories, and high in vitamin C


Featured Fruit
Peach
Peach offers a host of health benefits.
It provides a heft dose of vitamin A.
The following nutrient content descriptors for peach: fat-free, saturated fat-free, sodium-free, cholesterol-free, high in vitamin A and a good source of vitamin C.
Featured Fruit
Peach
Peach offers a host of health benefits.
It provides a heft dose of vitamin A.
The following nutrient content descriptors for peach: fat-free, saturated fat-free, sodium-free, cholesterol-free, high in vitamin A and a good source of vitamin C.
Featured Fruit
Alex, the Peach
Alex offers a host of health benefits.
It provides a heft dose of vitamin A.
The following nutrient content descriptors for Alex: fat-free, saturated fat-free, sodium-free, cholesterol-free, high in vitamin A and a good source of vitamin C.
Featured Fruit
Alex, the Peach
Alex offers a host of health benefits.
It provides a heft dose of vitamin A.
The following nutrient content descriptors for Alex: fat-free, saturated fat-free, sodium-free, cholesterol-free, high in vitamin A and a good source of vitamin C.

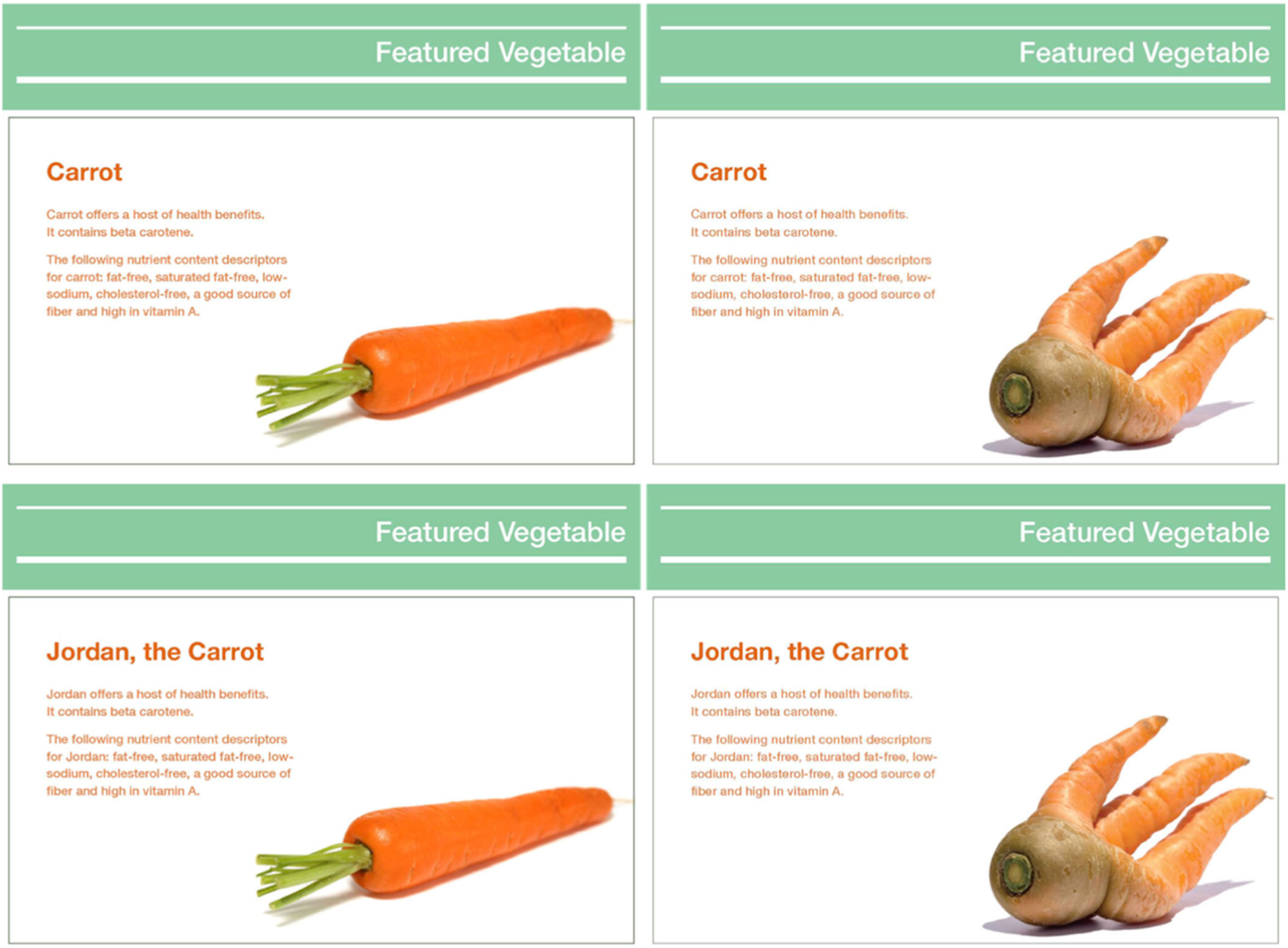
Study 3 Stimuli

Large Corporate Farm versus Small Local Farm

Large Corporate Farm vs. Small Local Farm


no-naming
vs.
anthropomorphic naming
Featured Vegetable
Eggplant
Eggplant offers a host of health benefits. It contains antioxidants.
The following nutrient content descriptors for eggplant: fat-free, saturated fat-free, sodium-free, cholesterol-free and low in calories.
Featured Vegetable
Eggplant
Eggplant offers a host of health benefits. It contains antioxidants.
The following nutrient content descriptors for eggplant: fat-free, saturated fat-free, sodium-free, cholesterol-free and low in calories.
Featured Vegetable
Taylor, the Eggplant
Taylor offers a host of health benefits. It contains antioxidants.
The following nutrient content descriptors for Taylor: fat-free, saturated fat-free, sodium-free, cholesterol-free and low in calories.
Featured Vegetable
Taylor, the Eggplant
Taylor offers a host of health benefits. It contains antioxidants.
The following nutrient content descriptors for Taylor: fat-free, saturated fat-free, sodium-free, cholesterol-free and low in calories.


Featured Vegetable
Tomato
Tomato offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for tomato: low-fat, saturated fat-free, sodium-free, cholesterol-free, low in calories, a good source of vitamin A and high in vitamin C.
Featured Vegetable
Tomato
Tomato offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for tomato: low-fat, saturated fat-free, sodium-free, cholesterol-free, low in calories, a good source of vitamin A and high in vitamin C.
Featured Vegetable
Alex, the Tomato
Alex offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for Alex: low-fat, saturated fat-free, sodium-free, cholesterol-free, low in calories, a good source of vitamin A and high in vitamin C.
Featured Vegetable
Alex, the Tomato
Alex offers a host of health benefits.
It provides a heft dose of vitamin C.
The following nutrient content descriptors for Alex: low-fat, saturated fat-free, sodium-free, cholesterol-free, low in calories, a good source of vitamin A and high in vitamin C.



Featured Vegetable
Carrot
Carrot offers a host of health benefits. It contains beta carotene.
The following nutrient content descriptors for carrot: fat-free, saturated fat-free, low-sodium, cholesterol-free, a good source of fiber and high in vitamin A.
Featured Vegetable
Carrot
Carrot offers a host of health benefits. It contains beta carotene.
The following nutrient content descriptors for carrot: fat-free, saturated fat-free, low-sodium, cholesterol-free, a good source of fiber and high in vitamin A.
Featured Vegetable
Jordan, the Carrot
Jordan offers a host of health benefits. It contains beta carotene.
The following nutrient content descriptors for Jordan: fat-free, saturated fat-free, low-sodium, cholesterol-free, a good source of fiber and high in vitamin A.
Featured Vegetable
Jordan, the Carrot
Jordan offers a host of health benefits. It contains beta carotene.
The following nutrient content descriptors for Jordan: fat-free, saturated fat-free, low-sodium, cholesterol-free, a good source of fiber and high in vitamin A.